The Core Lemma via arch telescoping: a proven reduction to a local forest inequality

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Abstract

We sharpen the reduction of the Core Lemma $\mathsf{mincost}(\lambda) \geq n(\lambda)$ (the open min-degree lower bound for $\chi^\lambda(\Omega)$). In any feasible closed walk along the staircase word the swaps form a nested family of equal-comparator *arches* whose forest is a union of chains (verified $n \leq 7$). We prove a *telescoping identity* cost $= D(U_0) + \sum_a \Delta_a$ with Δ_a local, and a *refined per-arch inequality* relating Δ_a to the D-jump ΔD_a, both unconditionally. This reduces the Core Lemma to a purely local *Forest Inequality*; we show the per-arch inequality alone is insufficient and isolate the single missing ingredient as a parent–child realizability constraint.

1 Setup

$\lambda \vdash n$, rows/columns 0-indexed, $\mathrm{cont}(r, c) = c - r$, $\mathrm{cont}_T(v)$ the content of the cell of v in $T \in \mathrm{SYT}(\lambda)$; $\delta_T(i) = [\mathrm{cont}_T(i{+}1) < \mathrm{cont}_T(i)]$. The staircase word $\mathbf{w}_0 = (1)(2\,1)\cdots(n{-}1\cdots 1)$ has *block* $b = (b, \ldots, 1)$, so comparator i occurs once in each block $b \geq i$. A closed walk takes one STAY/SWAP step per letter; SWAP at i sends $U \mapsto s_i U$ (swap cells of labels $i, i{+}1$), legal iff $|d_U(i)| \geq 2$ with $d_U(i) = \mathrm{cont}_U(i{+}1) - \mathrm{cont}_U(i)$; *feasible* iff no before-state has $d = -1$. Cost cost $= \sum_k \delta_{U_k}(i_k)$, and $D(T) = \sum_i (n - i)\delta_T(i)$ is the all-stay cost. We cite the proved **all-stay lemma**: $D(T) \geq n(\lambda)$ for every T.

2 Structural Lemma (verified $n \leq 7$)

Lemma 1. *In every feasible closed walk:* (a) *the swaps match into nested* arches*, each a pair of swaps at one comparator i opening at a block b_1 and closing at $b_2 > b_1$ (equivalently the swap subword is stack-reducible by cancelling stack-adjacent equal comparators);* (b) *the nesting forest is a union of chains (each arch has ≤ 1 child);* (c) ≤ 1 *root for $n \leq 6$, ≤ 2 for $n = 7$.*

Status: exhaustively verified for all feasible closed walks $n \leq 6$ and a broad $n = 7$ sample; zero violations. Proof open.

Fix an arch a: comparator i, blocks $b_1 < b_2$, boundary state U (just before the opening swap), inner $V = s_i U$. Write $w := b_2 - b_1 \geq 1$, $ni := n - i$. As $b_1 \geq i,\ b_2 \leq n - 1$,

$$1 \leq w \leq ni - 1. \tag{1}$$

For an *innermost* arch the region strictly between its swaps is all-stay at V.

3 Per-arch algebra (proved)

With $x_j := \mathrm{cont}_U(j)$ set $\gamma = \delta_V(i) - \delta_U(i)$, $\alpha = \delta_V(i-1) - \delta_U(i-1)$, $\beta = \delta_V(i+1) - \delta_U(i+1)$ (and $\alpha = 0$ if $i = 1$, $\beta = 0$ if $i + 1 = n$), $\sigma = \gamma + \alpha + \beta$.

Lemma 2 (sign rule and σ-linkage)**.** $\gamma \in \{\pm 1\}$ *with* $\gamma = +1 \iff x_i < x_{i+1}$*;* $\alpha\gamma \le 0$, $\beta\gamma \le 0$*;* $\sigma \in \{-1, 0, 1\}$*;* $\sigma = \pm 1 \Rightarrow \alpha - \beta = 0$*;* $\sigma = 0 \Rightarrow |\alpha - \beta| = 1$*.*

Proof. Legality gives $x_i \ne x_{i+1}$; toggling labels $i, i{+}1$ flips the descent at i, so $\delta_V(i) = 1 - \delta_U(i)$ and $\gamma = 1 - 2\delta_U(i) = \pm 1$, $\gamma = +1 \iff x_i < x_{i+1}$. In V, label i has content x_{i+1} and label $i{+}1$ has content x_i; x_{i-1}, x_{i+2} unchanged. Hence $\alpha = [x_{i+1} < x_{i-1}] - [x_i < x_{i-1}]$ and $\beta = [x_{i+2} < x_i] - [x_{i+2} < x_{i+1}]$. If $\gamma = +1$ ($x_{i+1} > x_i$) both are ≤ 0; if $\gamma = -1$ both are ≥ 0. Thus $\alpha, \beta \in \{0, -\gamma\}$, giving the sign rule and $\sigma \in \{-1, 0, 1\}$. Enumerating the sign-consistent (α, β) shows $\sigma = \pm 1$ forces $\alpha = \beta$ and $\sigma = 0$ forces exactly one of them nonzero. □

(Brute force: all 3800 legal swaps over every SYT, $n = 3..8$, satisfy Lemma 2.)

Lemma 3 (D-jump)**.** $\Delta D_a := D(V) - D(U) = ni \cdot \sigma + (\alpha - \beta)$.

Proof. U, V differ only at comparators $i{-}1, i, i{+}1$: $D(V) - D(U) = (ni{+}1)\alpha + ni\gamma + (ni{-}1)\beta = ni\sigma + (\alpha - \beta)$. □

Lemma 4 (local cost-change)**.** *Removing an innermost arch (inner segment re-set to all-stay at U, two swaps dropped) changes the cost by exactly $\Delta_a = w\,\sigma = (b_2 - b_1)(\gamma + \alpha + \beta)$.*

Proof. The region strictly between the two comparator-i swaps contains $b_2 - b_1 - 1$ occurrences of i and $b_2 - b_1$ each of $i{-}1, i{+}1$ (tail of block b_1, full blocks $b_1{+}1..b_2{-}1$, head of block b_2). The cost difference (at V minus at U) is γ, α, β at $i, i{-}1, i{+}1$; the two swaps minus two all-stays at i net γ. Summing: $\Delta_a = \gamma + (b_2{-}b_1{-}1)\gamma + (b_2{-}b_1)(\alpha + \beta) = (b_2 - b_1)\sigma$. □

Corollary 5 (telescoping)**.** *Given Lemma 1(a),* $\mathrm{cost}(W) = D(U_0) + \sum_{arches} \Delta_a$*.*

Theorem 6 (refined per-arch inequality)**.** $\Delta D_a \ne 0$*;* $\Delta D_a < 0 \Rightarrow \Delta_a \ge \Delta D_a + 1$*;* $\Delta D_a > 0 \Rightarrow \Delta_a \ge 0$*.*

Proof. $\Delta_a = w\sigma$, $\Delta D_a = ni\,\sigma + (\alpha - \beta)$, $\sigma \in \{-1, 0, 1\}$, $1 \le w \le ni{-}1$. If $\sigma = 0$: $\Delta D_a = \alpha - \beta = \pm 1$, $\Delta_a = 0$, and $0 \ge \Delta D_a + 1$ when $\Delta D_a = -1$, $0 \ge 0$ when $\Delta D_a = +1$. If $\sigma = +1$: $\Delta D_a = ni > 0$, $\Delta_a = w \ge 0$. If $\sigma = -1$: $\Delta D_a = -ni < 0$, $\Delta_a = -w \ge -(ni - 1) = \Delta D_a + 1$ by (1). □

4 Reduction to the Forest Inequality

By Lemma 1 the arches form a forest of chains. With child $c(a)$: $T(a) = \Delta_a + T(c(a))$, $M(a) = \Delta D_a + \min(0, M(c(a)))$ (and $T = \Delta, M = \Delta D$ at a leaf). Then $\min_t D(U_t) = D(U_0) + \mu$ where $\mu = \min(0, \min_{\text{roots } r} M(r))$, and by Corollary 5 $\mathrm{cost} - D(U_0) = \sum_r T(r)$. Hence, modulo Lemma 1, the Core Lemma is equivalent to the

$$\textbf{Forest Inequality:} \qquad \sum_{\text{roots } r} T(r) \;\ge\; \mu.$$

Verified with zero violations for all feasible closed walks $n \le 7$.

5 The parent–child realizability constraint

The reduction so far uses only the per-arch data; the per-arch inequality alone is *insufficient*. Indeed a length-2 chain with parent (σ=1, ni=3, w=1), i.e. $(\Delta, \Delta D) = (1, 3)$, and child ($\sigma$= −1, ni=4, w=3), i.e. $(\Delta, \Delta D) = (-3, -4)$, satisfies Theorem 6 and (1) but gives $\sum \Delta = -2 < \mu = -1$. No feasible walk realizes it. The missing facts are *parent–child realizability constraints* coming from two coupling principles.

Boundary identity. In a chain, between the parent's opening swap and the child's opening swap every step is a STAY (the child is the unique nested arch), and STAYs fix the tableau. Hence the child's boundary state equals the parent's inner state:

$$U_c = V_p = s_{i_p} U_p. \tag{2}$$

This couples *all* child data $(\sigma_c, ni_c, w_c, e_c)$ to the parent, since they are read off contents of $U_c = V_p$.

Lemma 7 (width coupling)**.** *For a parent arch a (comparator i_p, blocks $b_1^p < b_2^p$) and its child $c(a)$ (comparator i_c, blocks $b_1^c < b_2^c$), one has $w_{c(a)} \le w_a - 1$.*

Proof. Index each letter of $\mathbf{w}_0$ by (b, i) and order by step order $\prec$: $(b, i) \prec (b', i')$ iff $b < b'$, or $b = b'$ and $i > i'$ (within a block the comparators descend). Nesting gives $(b_1^p, i_p) \prec (b_1^c, i_c)$ and $(b_2^c, i_c) \prec (b_2^p, i_p)$. From the first, $b_1^c \ge b_1^p$, with equality only if $i_c < i_p$. From the second, $b_2^c \le b_2^p$, with equality only if $i_c > i_p$. The two equalities require $i_c < i_p$ and $i_c > i_p$ simultaneously, impossible; and if $i_c = i_p$ neither equality holds. So at least one inequality is strict, giving $w_{c(a)} = b_2^c - b_1^c \le (b_2^p - b_1^p) - 1 = w_a - 1$. □

Lemma **??** (verified $n \le 7$, tight) already excludes the fake chain above ($w_c = 3 \not\le w_p - 1 = 0$) and *every* length-2 violator. The residual, after imposing $w_c \le w_p - 1$, is a family of length-≥ 3 abstract chains stacking two (σ=0, e=+1) arches over a (σ= − 1, ni=2, w=1) leaf. These are eliminated by two further consequences of (**??**), each verified with zero violations over all feasible closed walks $n \le 7$:

- **(R)** a σ=0 child of a σ=0 parent inherits e: $e_c = e_p$;
- **(S2)** a (σ=0, e=+1) arch whose parent has σ=0 is a leaf (equivalently, no chain contains $(\sigma{=}0) \to (\sigma{=}0, e{=}{+}1) \to (\cdot)$).

Together with Theorem 6 and Lemma **??**, (R)+(S2) close the Forest Inequality on all enumerated abstract chains.

6 Status

Proved unconditionally: Lemmas 2–4, Corollary 5 (given Lemma 1(a)), Theorem 6, and the *width-coupling* Lemma **??** (from the boundary identity (**??**) and the step order of $\mathbf{w}_0$). Lemma **??** alone removes the fake chain and all length-2 violators. **Verified $n \le 7$ (open):** Lemma 1, the content-coupling rules (R) and (S2), and the Forest Inequality. **Remaining gap:** derive (R) and (S2) from the boundary identity $U_c = V_p$ via the content arithmetic of $V_p = s_{i_p} U_p$ (e.g. how the descent flips at $i_p \pm 1$ propagate to the child comparator i_c), plus a slack-absorption bound dispatching the ≤ 2 roots. With Lemma **??** the gap is now a *content*-coupling statement, not a width one: the w-geometry is fully proved.